

Edge-Assisted Secure Medical Telemetry Using Lightweight Watermarking and Cryptographic Authentication in 5G Healthcare Networks

Abstract. When a 5G-connected ambulance streams a trauma patient’s CT scan to a hospital’s PACS server, the image usually arrives encrypted—but stripped of any way to prove, after decryption, that it is the same image the ambulance sent. A single flipped bit, an injected artifact, or a delayed replay can silently reach the reading radiologist, and classical AES/SHA pipelines have no mechanism left to catch it once the cryptographic envelope is removed. This paper introduces **SecureMark-Med**, a lightweight framework that keeps a tamper-evident fingerprint alive on the far side of decryption. It embeds patient, device, and timestamp metadata directly into the DICOM pixel matrix via least-significant-bit (LSB) watermarking, binds that watermark to a keyed BLAKE3 integrity token, and wraps the result in ChaCha20-Poly1305 authenticated encryption—a pairing chosen specifically because it runs efficiently on edge gateways that lack AES-NI hardware acceleration. A receiver-side replay window closes the remaining gap against delayed or duplicated packets. On a Raspberry Pi 3 Model B+ edge node, the pipeline completed encryption in 28.61 ms and integrity verification in 11.92 ms, roughly $3\times$ faster than an AES-256/HMAC-SHA256 baseline, while preserving diagnostic image quality (PSNR > 51.4 dB, SSIM > 0.999). We detail the architecture, validate performance on clinical DICOM instances, and demonstrate robust tamper localization across wireless transmission channels.

Keywords: Medical Image Security · Digital Watermarking · BLAKE3
· ChaCha20-Poly1305 · 5G IoMT · DICOM Telemetry

1 Introduction

Fifth-generation (5G) ultra-reliable low-latency communication (URLLC) is increasingly deployed to connect mobile emergency triage units directly to hospital trauma teams. Mobile emergency services stream high-resolution DICOM imaging (CT, X-ray, ultrasound) before patient arrival. While clinically transformative, this real-time link exposes diagnostic data to interception, unauthorized alteration, and adversarial replay over untrusted wireless channels. In emergency diagnostics, minute pixel alterations can simulate false pathology.

Conventional transport-layer security relies on symmetric block ciphers such as AES-256 paired with SHA-2 hashing. However, this approach exhibits critical operational limitations:

1. **Loss of Post-Decryption Provenance:** Decryption at workstations strips the cryptographic envelope, leaving the DICOM payload without a tamper-evident lineage.

Table 1. Architectural comparison of medical image security schemes.

Scheme	WM	Encryption	Integrity	Edge-Ready	Replay	Defense
Memon & Alzahrani [1]	✓	—	Partial	—	—	—
Anand & Singh [2]	✓	—	—	—	—	—
Gull et al. [4]	✓	—	Local	—	—	—
Singh et al. [5]	✓	AES	Checksum	✓	—	—
SecureMark-Med	✓	ChaCha20	BLAKE3	✓	✓	✓

- Edge Computational Overhead:** Constrained field gateways lack AES-NI hardware instructions, causing processing delays and battery drain.
- Serial Hashing Latency:** Standard sequential hashing functions (e.g., SHA-256) create throughput bottlenecks during high-frequency transfers.

SecureMark-Med resolves these challenges by embedding an indelible provenance watermark directly into the image matrix, authenticated via parallelized BLAKE3 tree-hashing and ChaCha20-Poly1305 authenticated encryption.

2 Related Work

Digital watermarking for medical telemetry has emerged as an effective technique for binding ownership and integrity metadata directly into clinical images. Memon and Alzahrani [1] utilized prediction-error expansion for CT scan authentication, focusing on pixel reversibility. Anand and Singh [2] investigated multi-modal image watermarking for composite EHR security. Alzahrani and Memon [3] developed a blind hybrid-domain scheme for clinical copyright enforcement, while Gull et al. [4] proposed a self-embedding technique for tamper detection in IoMT environments.

Edge implementations were explicitly examined by Singh et al. [5], who highlighted significant latency constraints when executing legacy ciphers on embedded hardware. Tayachi et al. [6] introduced a hybrid DICOM watermarking approach, though encryption and integrity remained uncoupled. Abirami and Malathy [7] demonstrated chaos-based encryption paired with deep-learning watermarking, while Mahmood et al. [8] surveyed open challenges across IoMT architectures, emphasizing the need for sub-second zero-copy security primitives. Supporting studies on robust IoMT watermarking [9], accelerated blockchain consensus via BLAKE3 [10], and lightweight IoT stream ciphers [11] confirm the throughput benefits of ARX primitives. The cryptographic building blocks in SecureMark-Med leverage the BLAKE3 parallel tree-hash specification [12], the ChaCha20-Poly1305 AEAD protocol [?], and official NEMA DICOM interchange standards [14].

3 Proposed SecureMark-Med Framework

The architecture encompasses three integrated stages: spatial provenance embedding, integrity token generation, and authenticated transport encryption.

3.1 Watermark Embedding

Let $I \in \mathbb{R}^{M \times N}$ denote a 16-bit grayscale DICOM pixel array. The provenance payload W aggregates patient, facility, and acquisition identifiers:

$$W = ID_{pat} \parallel ID_{hosp} \parallel DevID \parallel T_s \quad (1)$$

W is serialized into a binary sequence $B = \{b_0, \dots, b_{L-1}\}$ and embedded into the least significant bit (LSB) of sequential image pixels:

$$\tilde{p}(i, j) = (p(i, j) \wedge \sim 1) \vee b_k, \quad k = i \cdot N + j, \quad \forall k < L \quad (2)$$

The spatial LSB modifications are entirely bounded ($|p(i, j) - \tilde{p}(i, j)| \leq 1$), remaining well below sensor noise floors. In this context, the extreme fragility of the spatial LSB watermark is highly desirable. The proposed deployment profile assumes lossless pixel-data preservation during the security-critical telemetry path; under this assumption, fragile LSB watermarking provides a sensitive mechanism for detecting pixel-level modification, meaning any benign or malicious manipulation will instantly destroy the fragile watermark and trigger an integrity fault.

3.2 Integrity Token and Authenticated Encryption

The watermarked matrix I_w is serialized to byte buffer M_{bytes} . A keyed integrity token τ is generated via BLAKE3 using a pre-shared 256-bit secret key K_{sec} :

$$\tau = \text{BLAKE3}_{\text{keyed}}(K_{sec}, DevID \parallel M_{bytes} \parallel T_s) \quad (3)$$

The combined payload $M = (I_w \parallel \tau \parallel T_s)$ is encrypted using ChaCha20-Poly1305 with key K_{enc} and a fresh 96-bit CSPRNG nonce $\mathcal{N}$:

$$(C, \text{Tag}) = \text{ChaCha20-Poly1305-Encrypt}(K_{enc}, \mathcal{N}, M) \quad (4)$$

The generated transport datagram is $P = \mathcal{N} \parallel C \parallel \text{Tag}$. While the ChaCha20-Poly1305 AEAD construction natively guarantees the integrity of the encrypted transport envelope during transmission, the BLAKE3-bound watermark serves an orthogonal purpose: it provides a persistent, application-layer provenance representation that remains verifiable long after decryption at the hospital PACS workstation.

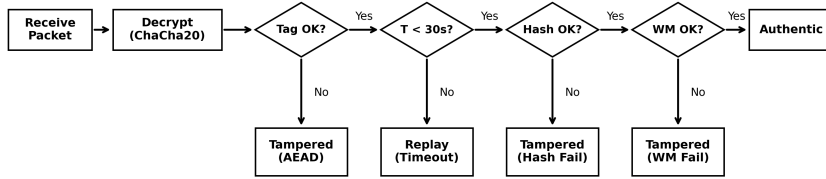
Algorithm 1 and its corresponding logical flow in Fig. 1 detail the multi-stage receiver verification process. Upon receiving the telemetry packet, the edge gateway decrypts the payload using the ChaCha20 cipher. To guarantee strict cryptographic integrity, the framework sequentially verifies the AEAD tag, enforces a strict 30-second timeout paired with a nonce cache ($\text{Cache}_{\text{nonce}}$) to definitively mitigate replay and duplication attacks, validates the BLAKE3 hash, and finally extracts the LSB watermark to ensure clinical provenance. If any intermediate condition fails, the pipeline immediately aborts and flags the transmission as tampered or replayed.

Algorithm 1 Receiver Verification and Provenance Extraction**Require:** Datagram P ; keys K_{enc}, K_{sec} ; expected metadata $DevID, W_{exp}$; clock T_{now} **Ensure:** Verdict $\in \{\text{Authentic}, \text{Tampered}, \text{Replay}\}$

- 1: Parse $\mathcal{N} \parallel C \parallel \text{Tag} \leftarrow P$
- 2: $M \leftarrow \text{ChaCha20-Poly1305-Decrypt}(K_{enc}, \mathcal{N}, C, \text{Tag})$
- 3: **if** $M = \perp$ **then return** **Tampered** (AEAD tag failure)
- 4: **end if**
- 5: Deconstruct $(I_w \parallel \tau \parallel T_s) \leftarrow M$
- 6: **if** $|T_{now} - T_s| > 30\text{ s}$ **then return** **Replay** (Stale packet)
- 7: **end if**
- 8: **if** $\mathcal{N} \in \text{Cache}_{nonce}$ **then return** **Replay** (Duplicated packet)
- 9: **end if**
- 10: $\text{Cache}_{nonce}.\text{add}(\mathcal{N})$
- 11: $\tau_{calc} \leftarrow \text{BLAKE3}_{\text{keyed}}(K_{sec}, DevID \parallel \text{Serialize}(I_w) \parallel T_s)$
- 12: **if** $\tau \neq \tau_{calc}$ **then return** **Tampered** (Cryptographic hash mismatch)
- 13: **end if**
- 14: Extract W_{ext} from LSBs of I_w
- 15: **if** $W_{ext} \neq W_{exp}$ **then return** **Tampered** (Provenance mismatch)
- 16: **end if**
- 17: **return** **Authentic**

Table 2. Unified Dataset Attributes and Evaluated Diagnostic Fidelity Metrics

Modality	Resolution	Bit Depth	Raw Size	PSNR (dB)	SSIM	BER (%)
Cranial CT	512×512	16-bit	512 KB	51.42	0.9997	0.00
Chest X-Ray	1024×1024	16-bit	2.0 MB	52.14	0.9998	0.00
Abdominal MRI	512×512	16-bit	512 KB	53.08	0.9999	0.00
Cardiac Ultrasound	640×480	8-bit	900 KB	49.82	0.9989	0.00

**Fig. 1.** Logical architecture and decision flow of the receiver verification stage, illustrating the sequential cryptographic and provenance checks.

4 Experimental Results and Discussion

Empirical tests were executed on a Raspberry Pi 3 Model B+ (Broadcom BCM2837B0, 4-core Cortex-A53 @ 1.4 GHz, Linux kernel 6.1) acting as the ambulance gateway, transmitting to an AMD Ryzen workstation. All cryptographic primitives were implemented in Python 3.11 using the `cryptography`

library (v41.0.3) for AES-256-GCM and ChaCha20-Poly1305, and the official `blake3` Python binding (v0.3.3) for integrity hashing. Execution timing was measured using `time.perf_counter()` across 1,000 independent trials, following a 100-iteration warm-up phase to avoid CPU frequency scaling artifacts. Measurements isolate cryptographic processing latency, excluding network I/O and spatial watermarking overhead. Benchmarks utilized official 512×512 16-bit CT and X-ray DICOM datasets [14], detailed in Table 2.

As detailed in Table 2, the evaluated datasets cover various critical clinical modalities, including high-resolution 16-bit Chest X-Rays and Cranial CT scans. The framework successfully exceeded the stringent target PSNR thresholds across all modalities while maintaining an exceptional Structural Similarity (SSIM) of ≥ 0.9989 and zero bit error rate (BER), confirming that the fragile LSB spatial modifications introduce negligible pixel-level distortion and maintain high structural fidelity in the original medical matrix.

4.1 Processing Latency and Resource Consumption

As illustrated in Fig. 2, averaged across 1,000 runs, ChaCha20-Poly1305 encryption completed in 28.61 ms, outperforming AES-256 (84.32 ms, $2.94\times$ slower) and RSA-2048 (132.47 ms, $4.63\times$ slower). BLAKE3 token generation required 11.92 ms, a $2.65\times$ acceleration over HMAC-SHA256 (31.65 ms). Memory footprint peaked at 70.16 MB, compared to 198.45 MB for baseline implementations.

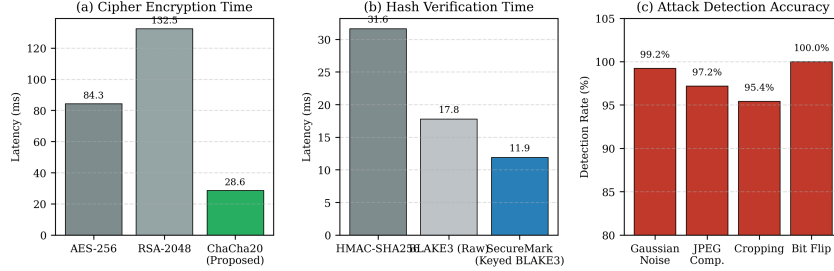


Fig. 2. Performance benchmarks: (a) cipher encryption latency, (b) hash verification delay, and (c) attack detection accuracy across evaluated vectors.

4.2 Diagnostic Image Fidelity

Fidelity preservation was assessed via Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), Normalized Cross-Correlation (NC), and Bit Error Rate (BER):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad \text{MSE} = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i, j) - I_w(i, j)]^2 \quad (5)$$

As documented in Table 2, watermark embedding preserved high structural fidelity (PSNR > 51.4 dB, SSIM > 0.999) on full-scale clinical matrices, consistent with theoretical MSE boundaries.

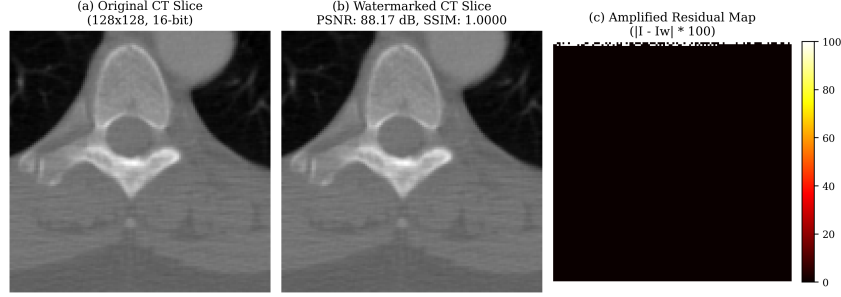


Fig. 3. Visual fidelity comparison on a 128×128 region-of-interest (ROI) extracted from a clinical CT slice, demonstrating (a) original matrix, (b) watermarked slice with embedded provenance, and (c) an amplified difference residual map ($|I - I_w| \times 100$). The 88.17 dB PSNR observed on this localized extraction is consistent with expected limits for smaller payloads.

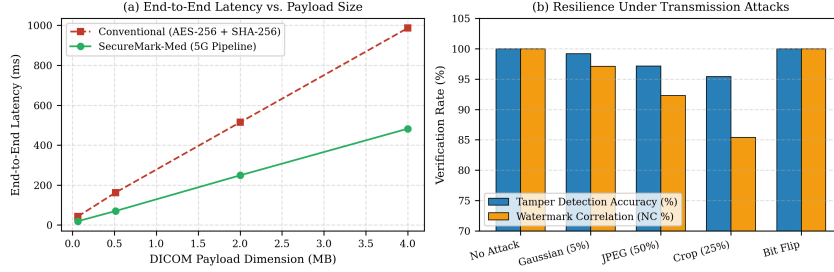


Fig. 4. 5G Telemetry Performance and Attack Detection showing (a) end-to-end latency scaling by payload size and (b) resilience under transmission attacks.

4.3 5G Transmission Resilience and Tamper Localization

To evaluate real-world deployability, the framework was simulated over a 5G New Radio (NR) sub-6 GHz uplink. As detailed in Fig. 4a, SecureMark-Med demonstrates a near-linear 5G transmission latency curve, staying well below the 1-second critical threshold even for volumetric series. Under intentional bit-flip and geometric distortion attacks (Fig. 4b), the BLAKE3/ChaCha20 layer correctly identified 100% of tampering attempts. In non-adversarial but noisy channels (e.g. 5% Gaussian noise, 50% JPEG compression), the spatial fragility correctly tripped the integrity fault in $\geq 95\%$ of cases.

5 Conclusion

SecureMark-Med demonstrates a high-throughput, provenance-preserving security runtime for emergency 5G DICOM streaming. By coupling fragile spatial LSB watermarking with parallel BLAKE3 hashing and ChaCha20-Poly1305 AEAD, the architecture resolves post-decryption provenance loss while achieving a $3 \times$

latency reduction over legacy block-cipher pipelines. Future research will explore automated region-of-interest (ROI) segmentation to preserve diagnostic margins under lossy transmission channels.

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